

Yeditepe University
Industrial Engineering Department

ISE 457
Forecasting Techniques
SPRING 2024

PROJECT GUIDELINES

Learning Objectives

This project provides an opportunity for you to apply the skills and knowledge gained in the forecasting course to a practical problem, i.e., **to forecast a real time series using ETS and ARIMA models.**

Timeline

This project will be done in **teams of 4 or 5 students (NOT ≤ 3 nor ≥ 6 students).** For graduate students (i.e., ESYE 547), please consult the instructor.

- **FORMING TEAMS:** You should form your teams by **25 April 2024, Friday, 23:59.** Otherwise you will be randomly assigned to a **EXISTING/NEW** team. Fill out the google sheet in this link.
- **PROJECT REPORT** is due by **31 May 2024, Friday, 23:59,** through *YULEARN* submission (failure to submit the report would result in a 0% mark for the overall project, no exception!). The report should be **no more than 4000 words of prose** (i.e., around **15 single spaced pages**, and no limit on the amount of code). See next section for evaluation criteria. You **MUST follow the guidelines in the last section** for preparing your report.

Data

The data are monthly measures of retail trade volume of a arbitray country. The data files (`all_data_upto202212.csv` and `data_202301_202402.csv`) are provided in YULEARN under "Project" topic. Each team will be using a different time series, using one of the team member's student ID number as follows:

```
get_my_data2 <- function(student_id) {
  set.seed(student_id)
  all_data <- read.csv("all_data_upto202212.csv")
  data_2023to2024 <- read.csv("data_202301_202402.csv")
  all_data <- all_data |> mutate(Month = yearmonth(Month)) |> as_tsibble(index = Month, key
= c(State,Industry,Series.ID))
  data_2023to2024 <- data_2023to2024 |> mutate(Month = yearmonth(Month)) |> as_tsibble(index
= Month, key = c(State,Industry,Series.ID))
  while(TRUE) {
    retail <- filter(all_data, Series.ID == sample(Series.ID, 1))
    if(!any(is.na(fill_gaps(retail)$Turnover)))

    retail2023to2024 <- filter(data_2023to2024, Series.ID == sample(Series.ID, 1))
    if(!any(is.na(fill_gaps(retail2023to2024)$Turnover)))
    return(list(retail, retail2023to2024))
  }
}
# Replace the 12345678 argument with your student ID
retail<- get_my_data2(12345678)[[1]]
retail2023to2024<- get_my_data2(12345678)[[2]]
```

Report:

You should produce forecasts of the series using ETS and ARIMA models. Write a report in Markdown or Rmarkdown format of your analysis explaining carefully what you have done and why you have done it. Your report should include the following elements:

- A discussion of the statistical features of the original data, including the effect of COVID-19 on your series. Include a time series decomposition analysis (STL) with explanations as well.
- Explanation of transformations and differencing used. You should use a unit-root test as part of the discussion.
- A description of the methodology used to create a short-list of appropriate ARIMA models and ETS models. Include discussion of AIC values as well as results from applying the models to a test-set consisting of the last 24 months of the data provided.
- Choose one ARIMA model and one ETS model based on this analysis and show parameter estimates, residual diagnostics, forecasts and prediction intervals for both models. Diagnostic checking for both models should include ACF graphs and the Ljung-Box test.
- Comparison of the results from each of your preferred models. Which method do you think gives the better forecasts? Explain with reference to the test-set.
- Apply your two chosen models to the full data set, re-estimating the parameters but not changing the model structure. Produce out-of-sample point forecasts and 80% prediction intervals for each model for two years past the end of the data provided.
- Compare your forecasts with the actual numbers (i.e., `retail2023to2024` variable at the end of the code on page 1). How well did you do?
- A discussion of benefits and limitations of the models for your data.
- Graphs should be properly labelled, including appropriate units of measurement.

Evaluation of Project Reports

Your project reports will be assessed based on the following criteria:

1. Does the project clearly **describe the data**?
2. Does the report clearly explain the work that was done (i.e., **conclusions and your main findings**)? Provide detailed explanations of your code and methodology to facilitate understanding by others.
3. Does the report use techniques presented in the course (or clearly related to topics covered in the course) **to understand and analyze the data as well as to forecast**?
4. Is the report **concise and clear** (your ability to explain technical material well)?
5. Is quality of **writing (use of graphical representations)** acceptable? Create visualizations to illustrate key findings and insights from your analysis.
6. **Does each member of the team has equal contribution (very important)?**

General Guidelines for Project Report

- The project report must be written using the **MARKDOWN** language (i.e., **.ipynb files in Google Colab or R markdown files with .rmd extensions are accepted, no other format!!!**)
 - For markdown, you can use **Google Colab .ipynb notebooks**.
 - For R markdown, see this link. You can use **RStudio Cloud**.
 - Also see tutorials in for the report formats
 - * kaggle notebook (link)
 - * google colab notebook (link) OR
 - * nbviewer (link) link to a Jupyter notebook OR Rstudio
 - **NO Microsoft WORD FILE are accepted (just markdown language)!**
- The notebook should still be written as a traditional report, just one that makes reproducing your report much easier for anyone who is interested in diving more deeply into your work. It should be readable as a narrative explaining your project without requiring the reader to run any code.
- **The final report has a constraint of 4000 words of prose (i.e., around 15 pages), and no limit on the amount of code.** However, if you develop very large code blocks as part of the project, you should include them in a separate R file. Your notebook should be executable in cell order.
- The project report should have the **name of all team members**.
- Give references to the sources you use and type all formulas using proper format (i.e., in markdown language you can use LaTeX).
- Keep the length of your reports reasonable and as indicated.
- Include **a title info** with team members' names, student numbers, and date (and any other relevant information) and a **table of contents (TOC)**, see this link.
- Write "impersonally" and therefore never use "I", "we" or "my team".
- Be clear and concise in the way you write. Short sentences and paragraphs are better than long rambling ones. However, do not make the sentences too short or your report will look like a series of notes. It is a question of balance.
- Good grammar and punctuation; correct sentence and paragraph construction are all "a must". Your report should be free of spelling mistakes and grammatical errors. Use Language tools before submission.
- Use appropriate and clearly labeled tables, charts, graphs, etc. (these often convey information more vividly than lengthy text.) However, you must ensure that you make a reference to them; that you explain what they show; and that they are fully integrated into the report.
- Ensure that all sources such as quotes and statistics are properly referenced. Use the APA or Harvard Manual of styles for referencing, or one of your choice. You can use footnotes or endnotes for the references. If you are not sure how to cite, refer to the "Harvard System of Referencing Guide" (available at this link). The entire report should be written in your own words; verbatim (using exactly the same words) or paraphrasing (restating) the original sentences are not acceptable (i.e., never copy and paste from your references as **Turnitin/SafeAssign** will be used to check your reports against **plagiarism!**)
- Proofread the final version carefully to eliminate any typographical mistakes. Spell check alone may not be sufficient to clean up the report.
- Submit a copy of your project **report** using *YULEARN* submission (this will ensure reproducibility of your report and code) as:

- **either a .ipynb or a .rmd file** (that includes all your main text and codes, written in markdown language)
 - **a .pdf file** (print your .ipynb files or knit your .rmd file as an html file and then print it as a pdf file using any web browser (see this link for converting .ipynb files to .pdf))
 - Any other file you may require to submit (**optional**).
- Note that **only one of the team members should submit (on behalf of the team)**. If different members submit multiple files, the last file uploaded will be available to me (the rest of the files are overwritten).
 - FOLLOWING RULES APPLY FOR EACH TEAM MEMBER:
 - It is the responsibility of each team member to contact each other.
 - Arrange regular meetings for your project.
 - All team members should equally contribute to the project .
 - The team can reject (by **MAJORITY RULE**) a team member WHO DOES NOT CONTRIBUTE. This rejected team member's name/student ID should not appear in the FINAL REPORT and she/he will get a 0 mark for the project (no exceptions, no make-up).

A note on plagiarism

Plagiarism, the unattributed copying of ideas, prose, pictures, program code, or any product of human effort, is forbidden. For this project, anything that you include as part of the written code, prose, or figures must be your own work, and cannot be copied from elsewhere (i.e., even with citation).